Figure A3 — L-Katyusha: exact one-step contraction of the Lyapunov function versus Lemma 4.5, and tightness of the variance bound

hard instance (n=16, m=64, κ=10⁴)

logistic (n=1000, d=50, κ=10³, α=1)

logistic (n=1000, d=50, κ=10⁴, α=1)

Lyapunov $\Psi_k = Z_k + Y_k + W_k$

$E[\Psi_k]$ (mean over seeds)

Lemma 4.5 bound $(1-\rho)^k \Psi_0$

product of exact contractions

realized Ψ_k (seed 0)

contraction exponent per pass $(-n \cdot \log \rho)$

exact one-step contraction

Lemma 4.5 (paper parameters)

tightness ratio / share of Ψ

r_var: $E\|g - F'(x)\|^2 / (2L \cdot D_F(w,x))$

Z/Ψ

Y/Ψ

W/Ψ

effective passes (k / n)